

Students' Civic Online Reasoning: A National Portrait

Joel Breakstone¹, Mark Smith¹, Sam Wineburg¹,
Amie Rapaport², Jill Carle², Marshall Garland², and Anna Saavedra³

¹ Graduate School of Education, Stanford University

² Gibson Consulting Group, Austin, Texas

³ Dornesife Center for Economic and Social Research, University of Southern California

Author Note

This is a prepublication version of an article that has been accepted for publication in
Educational Researcher.

Correspondence concerning this article should be addressed to Joel Breakstone, Graduate
School of Education, Stanford University, 485 Lasuen Mall, Stanford, CA 94305. Email:
breakstone@stanford.edu

Abstract

Are today's students able to discern quality information from sham online? In the largest investigation of its kind, we administered an assessment to 3,446 high school students. Equipped with a live Internet connection, students responded to six constructed-response tasks. Students struggled on all of them. Asked to investigate a site claiming to "disseminate factual reports" on climate science, 96% never learned about the organization's ties to the fossil fuel industry. Two-thirds were unable to distinguish news stories from ads on a popular website's homepage. Over half believed that an anonymously posted Facebook video, shot in Russia, provided "strong evidence" of U.S. voter fraud. Instead of investigating the organization or group behind a site, students were often duped by weak signs of credibility: a website's "look", its top-level domain, the content on its About page, and the sheer quantity of information it provided. The study's sample reflected the demographic profile of high school students in the United States, and a multilevel regression model explored whether scores varied by student characteristics. Findings revealed differences in student abilities by grade level, self-reported grades, locality, socioeconomic status, race, maternal education, and free/reduced price lunch status. Taken together, these findings reveal an urgent need to prepare students to thrive in a world in which information flows ceaselessly across their screens.

Keywords: assessment, digital literacy, civics

Students' Civic Online Reasoning: A National Portrait

Teenagers swim in a sea of digital information. Outside of school and homework, they spend more than seven hours a day online (Rideout & Robb, 2019). Three quarters say they get news from social media “often” or “sometimes,” and nearly 70% do so through Internet videos (Dautrich, 2018). Given this reality, teenagers need to be able to sort fact from fiction online to understand issues that affect themselves, their communities, and their country. Are they equipped to do so? Evidence suggests that young people, like many Americans, struggle to evaluate online sources (Gasser et al., 2012; Hargittai et al., 2010; Lurie & Mustafaraj, 2018; McGrew et al., 2018; Pan et al., 2007). However, prior research has rarely asked students to contend with actual digital content and has never yielded a nation-wide perspective on students’ abilities. This study is the first to offer a national portrait across demographic groups of how students evaluated online sources.

Background

The health of a democracy depends on people’s ability to access reliable information (Hobbs, 2010; Mihailidis & Thevenin, 2013). If youth consume information without the ability to assess its credibility—unable to tell who is behind a cause and what their motives might be—they are easy marks for rogues of all stripes. On the Internet, traditional gatekeepers and hallmarks of authority are largely absent. It is thus imperative to democratic functioning that students know how to assess the quality of information on which they base their decisions (Lynch, 2017; Metzger et al., 2010).

Unfortunately, research has revealed shortcomings in students’ approaches to evaluating digital content. Students who said they would base evaluations on source information rarely did so when observed in real time (Hargittai et al., 2010). Students frequently ignored source

information (Bartlett & Miller, 2011; Barzilai & Zohar, 2012), focusing instead on the relevance of the information provided (Walraven et al., 2009) and basing their conclusions on a website's surface-level features (Coiro et al., 2015; McGrew et al., 2018). These findings may reflect deficiencies in how students are taught to judge the credibility of Internet sources. Many of the most widely used website evaluation materials—including those appearing on prestigious university websites—feature outdated strategies that can lead students dangerously astray (Breakstone, et al., 2018; Caulfield, 2016; Sullivan, 2019; Wineburg et al., 2020; Wineburg & Ziv, 2020).

Despite research pointing to students' lack of preparation, few tools exist to measure students' competency in the digital realm. Extant assessments rarely ask students to evaluate real sources. Instead, they fill out Likert-scale items (Ashley et al., 2013) or respond to scaffolded hypothetical scenarios (Leu et al., 2012). From such proxy measures, it is hard to know what students actually do online.

Limited assessment options spurred us to develop new measures. To do so, we first sought to identify expertise in the evaluation of online sources. We asked three groups to examine a series of websites: Stanford University students, university professors from five different institutions, and professional fact checkers from leading news outlets (Wineburg & McGrew, 2017). Students and academics often approached sites by “reading vertically,” starting at the top of their screen and proceeding to the bottom, pausing to comment on the site's prose and layout, the presence of references and contact information, and the site's top-level domain (dot-orgs good, dot-coms bad). Fact checkers took a radically different tack. Landing on an unfamiliar website, they clicked away from it and engaged in *lateral reading* (Wineburg & McGrew, 2019). They opened up new tabs across the horizontal axis of their browser and

searched for information about the organization or individual behind it. Only after surveying what other sites had to say did they return to their original starting place. Using this approach, fact checkers quickly sized up sites that masked their intent and backers. Students and academics, on the other hand, often dwelled on the original site, resulting in confusion about its real agenda or sponsor. Lateral reading allowed fact checkers to answer a crucial question: Who's behind the information? Their evaluations were guided by two others: (1) What's the evidence? (2) What do other sources say? Taken together, these strategies allowed fact checkers to better evaluate digital content than either academics or college students.

We refer to fact checkers' approach to searching for and evaluating social and political information as Civic Online Reasoning. Our purview is more targeted than the expansive fields of media and digital literacy, which can embrace topics ranging from cyberbullying to identity theft. It focuses squarely on how to sort fact from fiction online, a prerequisite for responsible civic engagement in the twenty-first century (Kahne et al., 2012; Mihailidis & Thevenin, 2013).

In 2015, our research team launched a two-year project to develop assessments that measured Civic Online Reasoning. We developed a range of tasks through an iterative process of prototyping, expert review, and piloting with students in classrooms across the country. We also used think-aloud protocols to ensure the assessments tapped the intended skills. These tasks asked students to evaluate real digital sources, ranging from Facebook videos to websites of advocacy groups pretending to be nonpartisan think tanks. Students from middle school to college struggled to perform basic evaluations of digital content (McGrew et al., 2018). Middle school students confused sponsored content with news stories. High school students trusted a photograph posted anonymously to a social media site. College students deemed a website about minimum wage policy as credible even though it was run by a public relations firm.

Despite widespread concern about digital misinformation, no prior research has gauged students' ability to evaluate online sources on a national scale. In this study, students completed a holistic assessment in which they went online to evaluate a range of sources. We analyzed data from 3,446 students, a sample that reflected the demographic profile of high school students in the United States. This approach yielded evidence of students' ability to evaluate online information and evidence of how student performance on the assessment varied by region, locality, grade level, race/ethnicity, gender, home language, grades in school, maternal education, and free/reduced lunch status.

Methods

Participating Districts

Using the 2016-2017 Common Core of Data from the National Center for Education Statistics (NCES), we compiled a list of all traditional public and charter schools in the United States that enrolled students through grade 12 (National Center for Education Statistics, n.d.-a). We categorized districts by region (Northeast, Midwest, South, West) and grouped them by locale within regions (urban, suburban, and rural/town), resulting in twelve separate categories.

We recruited at least one district in each of the twelve categories. We prioritized districts with racial and ethnic diversity, and we aimed to maximize the number of states represented.¹ The final sample included 16 districts in 14 states. Participating districts varied considerably in enrollment, racial and ethnic diversity, percentage of students receiving special education services, and proportion of schools eligible for Title I under the Elementary and Secondary Education Act (see Table 1).

¹ We calculated the multi-group racial/ethnic diversity for each school and district with Simpson's Diversity Index formula. We used Reardon and Townsend's (2018) Stata -seg- command. The measure reflects the probability that two students selected from a random draw in a given entity (e.g., a school or district) are members of the same racial/ethnic group.

Table 1*Characteristics of participating districts*

Region	Locale	Enrollment	Title I Schools	Hispanic	White	Black/ African American	Asian	Native American	Special Education	Limited English Prof
Northeast	Urban	22,359	89.7%	3.1%	32.7%	52.9%	3.6%	0.2%	21.9%	4.0%
Northeast	Suburban	15,878	0.0%	15.1%	73.1%	5.2%	4.0%	0.1%	17.2%	1.3%
Northeast	Rural	2,089	0.0%	1.8%	91.0%	1.6%	5.3%	0.3%	14.4%	1.0%
Northeast	Rural	2,034	83.3%	3.7%	88.9%	2.5%	1.3%	0.1%	17.7%	0.7%
Midwest	Urban	45,455	92.3%	14.6%	2.5%	80.7%	1.7%	0.2%	17.0%	12.3%
Midwest	Suburban	20,322	0.0%	3.1%	76.8%	4.0%	11.7%	0.2%	11.4%	2.2%
Midwest	Suburban	21,029	100.0%	8.7%	44.4%	23.7%	16.2%	0.3%	14.2%	10.5%
Midwest	Rural	5,122	100.0%	3.0%	89.3%	1.3%	1.6%	0.6%	15.5%	1.2%
Midwest	Rural	5,100	38.5%	4.5%	85.6%	1.8%	4.6%	1.2%	14.0%	2.1%
South	Urban	12,115	10.0%	15.7%	51.5%	11.2%	15.0%	0.3%	10.2%	10.4%
South	Suburban	15,942	60.0%	13.9%	58.5%	6.1%	3.1%	4.6%	17.2%	4.3%
South	Rural	427	100.0%	15.2%	71.9%	6.6%	0.9%	0.2%	10.5%	0.7%
South	Rural	2,486	100.0%	30.0%	66.3%	1.0%	0.6%	0.6%	9.5%	11.1%
West	Urban	29,682	9.6%	17.8%	73.8%	1.1%	3.2%	0.5%	*	5.1%
West	Suburban	4,074	0.0%	13.7%	76.1%	0.6%	1.3%	1.2%	9.5%	1.7%
West	Rural	6,579	68.2%	19.3%	65.6%	0.9%	1.2%	6.5%	14.5%	4.6%

Teacher Recruitment and Registration

We randomly selected three high schools (grades 9-12) to participate within each district; if a district had three or fewer high schools, all were included.² We invited all social studies teachers to participate at each of the selected schools. For each teacher who volunteered, we randomly selected two sections from their schedule and asked them to administer assessments in those sections. As a token of appreciation, we provided \$50 gift cards for each of the selected class sections that participated. Teachers could choose to administer the assessment in other sections as well.

Administration

² There were two exceptions. A Northeastern district had a large proportion of magnet schools. To ensure that we selected schools that represented the district, we stratified by magnet/traditional school type and then randomly selected two schools from each type. In a Midwestern district, some teachers from the three schools we selected also administered the assessment to students in a fourth school that we had not selected because those teachers taught in multiple schools.

The assessment was administered with an online survey platform. Participation was voluntary. Students completed the survey using Internet-connected computers during their regularly scheduled social studies class, and they were allowed to use the Internet at any time to answer the questions. The tasks appeared in the same order for all students. Students were not allowed to return to items after they had submitted their responses; however, no time limits were imposed. Teachers were allowed to provide technical assistance to students, but they could not help students evaluate sources, interpret questions, or compose responses.

Analytic Sample

We collected completed assessments from 6,445 students in 79 classrooms at 31 schools across 16 districts. To ensure our sample reflected the U.S. population of students by region and locale, we used the 2016-17 Common Core of Data from NCES to calculate the percentages of public school students in each geographical region of the U.S. (Midwest, Northeast, South, and West) and by locale within each region (rural, suburban, and urban). Next, we generated a random sample of 2,937 students that was proportionate to the distribution of students in each region and locale.

We examined the racial and ethnic composition of this sample and found that the proportions of Hispanic and Black/African American students were lower than the national population of public school students.³ To improve the representativeness of our sample, we returned to the set of completed assessments and added the remaining 509 students who identified themselves as Hispanic or Black/African American to the analytic sample regardless of

³ We created a composite variable for race/ethnicity based on students' responses to two questions: 1) Are you Hispanic or Latino? (yes/no), and 2) What is your race? (American Indian or Alaska Native, Asian, Black or African American, Native Hawaiian or other Pacific Islander, White, Other—please specify, Prefer not to say). In accordance with National Center for Education Statistics (n.d.-b) guidelines, all students who said "yes" to the first question were counted as Hispanic regardless their answer to the second question. Students who either selected "no" or did not respond to the first question were grouped by their response to the second question.

their district. Adding these students resulted in a final analytic sample of 3,446 students that better matched the racial/ethnic makeup of American public schools (see Table 2).

Table 2

Analytic Sample by Region, Locale, Race/Ethnicity, Gender, Grade Level, Free/Reduced Lunch Status, and Mother's Education

	% United States Enrollment ^a	Analytic Sample N	Analytic Sample	Difference Between Analytic Sample and % United States Enrollment	Analytic Sample without Missing
Region					
Northeast	16%	732	21%	5%*	-
Midwest	21%	852	24%	3%*	-
South	37%	1,084	31%	-6%*	-
West	25%	778	22%	-3%*	-
Locale					
Urban	31%	967	28%	-3%*	-
Suburban	42%	1,659	48%	6%*	-
Rural	27%	820	24%	-3%*	-
Race/Ethnicity					
Asian or Pacific Islander	5%	112	3%	-2%*	4%
American Indian/Alaska Native	1%	29	1%	0%	1%
Black or African American	15%	306	9%	-6%*	11%
Hispanic	26%	698	20%	-6%*	25%
Other	-	42	1%	-	2%
Two or more races	3%	224	7%	4%*	8%
White	49%	1,373	40%	-9%*	49%
Prefer not to say	-	28	1%	-	-
Missing	-	634	18%	-	-
Gender					
Male	52%	1,283	37%	-15%*	43%
Female	48%	1,443	42%	-6%*	49%
Non-binary/self-describe/prefer not to say	-	230	7%	-	8%
Missing	-	490	14%	-	-
Grade Level					
8 th	-	17	0%	-	1%
9 th	27%	820	24%	-3%*	28%
10 th	26%	491	14%	-12%*	17%
11 th	24%	867	25%	1%	29%
12 th	24%	761	22%	-2%*	26%
Missing	-	490	14%	-	-

	% United States Enrollment	Analytic Sample N	Analytic Sample	Difference Between Analytic Sample and % United States Enrollment	Analytic Sample without Missing
National School Lunch Program Status					
Eligible/Participant	52%	850	25%	-27%*	29%
Not eligible/Non-Participant	48%	1,475	43%	-5%*	50%
I don't know	-	533	15%	-	18%
Prefer not to say	-	84	2%	-	3%
Missing	-	504	15%	-	-
Mother's Education level^b					
Less than high school	-	222	6%	-	8%

* $p < .05$. Statistical significance of the difference between the analytic sample and the United States enrollment target for each student subpopulation was obtained from a one-sample difference in proportion tests.

^a U.S. enrollment data is from the National Center for Education Statistics Common Core of Data (2016-17).

^b Common Core of Data does not include data on mother's education for U.S. public school students.

Adding the Black/African American and Hispanic students changed the composition of the sample by region/locality, but the impact was small. Table 2 shows the geographic distribution of the final analytic sample. Students from suburban schools in the Northeast and Midwest were slightly overrepresented and students from the South were slightly underrepresented, but the overall sample reflected the geographic distribution of students in the United States. We also considered whether the sample reflected the self-reported grade level, gender, and free/reduced lunch eligibility (see Table 2).

Assessment Instrument

The assessment included six constructed-response tasks previously developed by our team (cf. McGrew et al., 2018). The tasks sampled a range of Civic Online Reasoning skills, such as lateral reading and evaluating evidence on social media. (Appendix A provides descriptions of the tasks; Appendix B includes the instrument.) For example, the Website Evaluation task (task 3 in Appendix B) assessed whether students could investigate who was behind a website. Students were provided a link to the homepage of co2science.org, an

organization whose About page states that they seek to “disseminate factual reports and sound commentary” on the environmental effects of carbon dioxide. The group is actually funded by fossil fuel companies, including ExxonMobil. Students were asked to judge whether the site was reliable, and a screen prompt reminded them that they could search online to answer.

A second task, Evaluating Evidence (task 1 in Appendix B), gauged students' ability to evaluate the trustworthiness of a social media post. The post presented students with a Facebook video from a user named “I on Flicks” and asked students if it constituted “strong evidence” of voter fraud during the 2016 Democratic primaries. The video included four silent clips of poll workers surreptitiously stuffing ballots into bins. Captions told viewers that the clips depicted Democratic primary elections in Illinois, Pennsylvania, and Arizona. The post accompanying the video read: “Have you ever noticed that the ONLY people caught committing voter fraud are Democrats?” None of this was true. The clips actually showed voter fraud in Russia, something easily ascertained by a brief web search.

After students completed the six tasks, they self-reported a range of demographic characteristics, including: age, grade, gender, race, ethnicity, primary language, typical grades in school, mother's education, and free/reduced lunch participation.

Rubric Design and Scoring Procedures

Rubric design was informed by prior research and development of Civic Online Reasoning assessments (McGrew et al., 2018; Wineburg et al., 2016). In *Mastery* responses, students evaluated online content by investigating the source of information, by interrogating the evidence presented, or by seeking out information from other reliable sources. *Emerging* responses were partially incorrect or did not fully articulate sound reasoning. *Beginning*

responses included incorrect or irrelevant strategies for evaluating online information. (See Appendix C for a sample rubric.)

Two independent raters scored student responses using a three-level rubric (Beginning - 0; Emerging - 1; Mastery - 2).⁴ Estimates of interrater reliability were high, with Kappa coefficients between .93 and .97 for each task (Task 1 = .95; Task 2 = .97; Task 3 = .94; Task 4 = .96; Task 5 = .94; Task 6 = .93).

Multivariate Regression Analysis

We used multivariate regression to explore the relationship between student characteristics and performance on the assessment. We first created an overall composite score for students ($n = 3,402$) across the six tasks. We dropped forty-four students in the analytic sample from the composite score analysis because none of their completed tasks could be scored (e.g., they indicated that they encountered technical problems on the tasks they completed). We then fit a multilevel regression model to explore differences in composite scores by a variety of student characteristics. Below, we outline how we constructed the composite score and specify the model used for the analysis.

Composite Score

The composite score was an average of students' scores on the items they completed.⁵ So, if a student completed all six items, then their composite score was the mean of their six scores; if they completed three items, their score was the mean of those three scores, and so on. The

⁴ Scores from the first rater were used for the regression analysis.

⁵ To ensure that combining the items into a single score was psychometrically sound, we conducted an exploratory factor analysis and found that the correlation patterns between items supported constructing a single scale. The eigenvalue for a single factor scale was 3.04, and a single factor model accounted for nearly 60% of the total variance in scores. We also estimated internal consistency and found that Cronbach's alpha was .58. Tasks 4 and 5 share the same stimulus, so we checked to ensure that scores for these two items were sufficiently independent to treat as separate items. Scores for these two items were only weakly correlated ($r = .19$), and crosstabs showed that there was little overlap in scores across the two questions.

average composite score ranged between 0 (a student received a score of 0 on all completed items) to 2 (a student received a score of 2 on all completed items).

Multivariate Analysis

We fit a single two-level hierarchical linear model (HLM), with random intercepts at the school level.⁶ We included students' self-reported grades as a measure of prior academic achievement because higher achieving students might score systematically higher. The "student grades" ranged from 1 (student reported they earned "mostly D's") to 7 (student reported "mostly A's"). We also anticipated that students in upper grades might perform better than students in lower grades, so we included grade level in the model.

Additionally, we included students' geographic region and locale to adjust for any systematic differences by student location. Finally, we included demographic variables that raw scores suggested might be relevant to student performance on the tasks, including race/ethnicity, gender, free/reduced lunch status, maternal education, and first language spoken at home.⁷ Appendix D describes these variables and how they were measured; Appendix E provides the percentage of students scoring Emerging or Mastery by task for each of the variables.

Results

Students struggled on all of the tasks. At least two thirds of student responses were scored as Beginning for each of the six tasks. On four of the six tasks, over 90% of students' responses were scored Beginning. Out of all the student responses to the six tasks, fewer than 3% received a Mastery score. Claims on Social Media 1 (task 4 in Appendix B) had the lowest proportion of

⁶ The intraclass correlation coefficient was .04, which indicated that most of the variability in the regression-adjusted composite scores was within schools and not between them. This may have been a byproduct of our study design, which prioritized schools with heterogeneous student populations. Diverse schools may have more within-school variability than schools with homogenous populations.

⁷ For each demographic subpopulation, we included dummy variables, with the largest group serving as the reference category. For instance, for student race/ethnicity, we included dummy variables for students classified as Asian, other, Hispanic, two or more races, and Black, with White respondents serving as the reference population. Table 4 provides coefficient estimates for each demographic subgroup included in the model.

Mastery responses, with less than 1% of students demonstrating a strong understanding of the online reasoning competencies measured. Evaluating Evidence had the highest proportion, with 8.7% earning a Mastery score. (See Table 3 for the distribution of scores by task.)

The mean composite score across all students was 0.17 out of a possible score of 2. (See Appendix F for the distribution of composite scores.) Fifty-eight percent of students had an average composite score of 0, indicating that they scored at the Beginning level on all of the tasks they completed. Of the 3,402 students in the analytic sample with a composite score, only 13 scored at the Mastery level on all six tasks (0.38%).

Table 3
Percent of Responses at Each Rubric Level by Task

	Evaluating Evidence	Webpage Comparison	Website Evaluation	Social Media 1	Social Media 2	Home Page Analysis
Beginning	77.4%	92.1%	96.8%	92.8%	95.4%	68.7%
Emerging	13.9%	6.7%	1.4%	6.9%	2.7%	26.6%
Mastery	8.7%	1.2%	1.8%	0.3%	1.9%	4.7%
Total (N)	3,119	2,678	2,994	2,431	2,431	2,969

Note. Some students in the analytic sample did not complete all tasks or did not provide responses that could be scored. Consequently, the total number of responses varied by task.

Website Evaluation Task

The Website Evaluation task, which presented students with a climate change website funded by fossil fuel companies, had the highest proportion of Beginning scores, with 96.8% of students receiving the lowest score. Instead of leaving the site, the vast majority of students remained glued to the site itself, drawn in by its top-level domain (.org), the recency of its updates, and the sheer quantity of information it included. A student from a suburban district in the Northeast wrote: “This page is a reliable source to obtain information from. You see in the URL that it ends in .org as opposed to .com.” This student was seemingly unaware that, unlike dot-gov or dot-edu sites, dot-org is an “open” domain; any individual or group can obtain one

without passing a character test or attesting to benevolent intent (Wineburg & Ziv, 2019).

Students also gave undue weight to the site's About page, not recognizing it as a curated portrait of how a group wished to be perceived. A student from the urban South wrote:

The "about us" tab does show that the organization is a non-profit dedicated to simply providing the research and stating what it may represent. In their position paper on CO₂, they provide evidence for both sides and state matters in a scientific manner. Therefore, I would say they are an unbiased and reliable source.

In contrast, strong student responses used other online sources to discover that co2science.org undermines climate change science on behalf of corporate funders. Establishing this was not hard but did require that students leave the site to investigate. Entering the group's name in their browsers and reading laterally called up a host of results that shed light on the group's agenda. Less than two percent of students did this. One student who did wrote:

co2science.org is not a reliable source because it has ties to large companies that want to purposefully mislead people when it comes to climate change. According to *USA TODAY*, Exxon has sponsored this nonprofit to pump out misleading information on climate change. According to the *Seattle Post-Intelligencer*, many of their scientists also have ties with energy lobbyists.

Evaluating Evidence Task

Over half of the students (52%) were taken in by the video purportedly showing ballot stuffing in the United States. They concluded that the video provided "strong evidence" of Democratic voter fraud in primary elections. A student from a rural district in the Midwest took the video at face value: "Yes, it shows video evidence of fraud in multiple different states at

multiple different times.” Another student from an urban district in the Northeast wrote, “Yes, because the video showed people entering fake votes into boxes.”

A quarter of the students rejected the video but did not provide a coherent rationale for doing so. Students focused on irrelevant features like the lack of audio, the video’s grainy quality, or insufficient narration from the post’s author. Others accepted the veracity of the video but ended up rejecting it because there were too few examples. A student from the South wrote, “The video only shows a few specific instances, and is not enough to blame the whole Democratic party of voter fraud.” Still others struggled to interpret what was depicted in the clips, ignoring the source and origin of the video altogether. A student explained:

I have no idea what is going on exactly and I see people put papers in the box, but there are cases where the votes are sneaked in. I am not sure this is a strong case of faking votes for a person, but it is a case for vote fraud.

Only 8.7% of students received a Mastery score by rejecting the video and providing a relevant explanation. These students understood they had no way of knowing if “I on Flicks” was a reliable source or whether the video depicted American voting fraud. A student from an urban district in the West questioned the video’s origin:

It does not provide enough evidence because we don't know who the people are in the video, and who the people are that made the video. There is very little context and information given, besides the place, and a little bit of commentary making claims to convince you.

Of the 3,119 responses on this task, only three students actually tracked down the source of the video. Again, doing so required little more than choosing relevant search terms (e.g., “video” and “Democratic ballot stuffing”) and scanning the search engine results page for sites like Snopes or

the BBC. A student from a suburban district in the West showed what could be learned by leaving the video and reading laterally:

Doing some research on the Internet, I found the exact same video as shown above.

However, the video which I found, which was presented by BBC News, showed that the cam footage had nothing to do with the 2016 Democratic primary elections. Instead, BBC presented the cam footage as that of the Russian 2016 election. According to the announcer of BBC News, the video showed that the people in Russia—not America—were committing voter fraud.

Such responses were few and far between.

Multivariate Results

Our regression analysis revealed significant differences in students' composite scores for a variety of demographic factors. Table 4 shows the regression coefficients and standard errors for each of the variables in our model. Adjusting for all other included variables, students who reported earning higher grades performed significantly better than students who reported earning lower grades. A one-step increase in self-reported grades (e.g., from “mostly B’s” to “mostly A’s and B’s”) was associated with a .03 improvement in average composite score (a noticeable improvement in a scale with a maximum of 2). Similarly, grade level was a significant predictor of student performance. Each increase in grade level (e.g., from sophomore to junior) was associated with a .03 increase in students' composite scores for students who were similar for all other reported characteristics.

Table 4
Results from Mixed Effects Regression Model

	Composite Score
Student grades	0.029 (.00)***
National School Lunch Program eligible	-0.044 (.02)**
Mother's education	0.014 (0.01)*
Grade level	0.029 (.07)**
Region (compared with Midwest)	
Northeast	-0.036 (.04)
South	0.009 (.03)
West	0.001 (.04)
Locality (compared with Rural)	
Suburban	0.043 (.03)
Urban	0.107 (.03)***
Gender (compared with Female)	
Male	0.017 (.01)
Non-binary/Prefer to self-describe	-0.020 (.03)
Home Language (compared with English)	
Other	-0.036 (.03)
Spanish	-0.012 (.02)
Race/Ethnicity (compared with White)	
Asian	0.019 (.03)
Black or African-American	-0.120 (.03)***
Other	-0.083 (.04)*
Two or more races	-0.039 (.02)
Hispanic	-0.014 (.18)
_cons	-0.157 (.05)**
Variance: School	.003 (.001)*
Intra-class correlation (ρ)	.044
Schools: n	31: 1,690
Wald X^2	87.59

*Note: * $p < .05$, ** $p < .01$, *** $p < .001$; Standard errors in parentheses.*

We also observed significant results for proxy measures of socioeconomic status. Maternal education was a significant predictor of average composite score. For every one-step increase in a mother's reported education level (e.g., from less than high school to high school, or from some college to a bachelor's degree), performance improved by about .01 points on average. Free/reduced price lunch status was also a significant predictor of performance, with

students eligible to receive free/reduced lunch performing .04 points lower, on average, than students not eligible, adjusting for all other variables in the model.

Our model also revealed significant differences by other characteristics. Urban students significantly outperformed their rural and suburban peers. Table 5 includes contrasts between each locale pair not included in the initial model. Adjusting for all other variables, urban students scored .11 points higher than rural students and .06 points higher than suburban students. Race/ethnicity was also a significant factor in our model. After adjusting for all other variables, Black/African American students in our sample had lower composite scores than students who were Asian/Pacific Islander (-.14 points), Hispanic (-.11 points), multiracial (-.08 points), or White (-.12 points). Table 5 includes contrasts for each race/ethnicity comparison not included in the initial model. Other variables were not significant predictors of student scores. Gender, geographic region, and home language were not significant predictors when adjusting for all other variables.

Table 5
Contrasts of Marginal Effects for Race/Ethnicity and Locale

	Contrast
Race/Ethnicity	
Black vs. Asian	-.139 (.04)**
Black vs. White	-.120 (.03) ***
Black vs. Other	-.037 (.04)
Black vs. Two or more races	-.081 (.03)**
Black vs. Hispanic	-.105 (.03)**
Locality	
Urban vs. Rural	0.107*(0.03)
Urban vs. Suburban	0.064*(0.03)

*Note: * $p < .05$, ** $p < .01$, *** $p < .001$; Standard errors in parentheses.*

Limitations

This study's sample was not randomly generated; as a result, it was not perfectly representative on all observable characteristics. Assembling a truly random, representative

sample of students in the United States for this type of assessment would have presented prohibitive administrative and financial barriers. In contrast to the National Assessment of Educational Progress (NAEP), we had no legislative mandate for administration and far fewer resources for data collection. In contrast to NAEP, we relied only on the goodwill of school districts to participate, negotiated research access with districts individually, and worked with technology administrators to ensure students' access to the open Internet during assessment administration. Additionally, it was a low-stakes assessment. Students were not required to participate, and their performance did not affect course grades. However, the same holds true for NAEP exams, which are held up as the "gold standard" of large-scale testing (National Center for Education Statistics, n.d.-c). Despite these limitations, this is the largest study of students' online reasoning skills to date in the United States.

Measuring students' Civic Online Reasoning at scale presented a variety of challenges. In the interest of ecological validity, we asked students to evaluate real online sources about social and political issues using the Internet. This openness stands in stark contrast to the tightly controlled environments typical of large-scale assessment. We could not control the sources that students encountered while searching online. Nor could we directly observe how students arrived at their answers. Giving students free rein online also meant greater heterogeneity in the responses collected, which made scoring more difficult. Going forward, researchers will need to weigh the benefits of asking students to engage with authentic sources against the measurement challenges that the accompanying freedom creates.

There are important dimensions of evaluating online sources that this study did not address. In particular, it did not account for students' political beliefs. Although the assessment deliberately included sources from across the political spectrum, students' beliefs may have

influenced their evaluations of sources about contentious social and political issues. The interaction between students' political orientations and their evaluations of digital sources obviously merits further study.

Discussion

The results of this study should give pause to anyone who cares about the future of democracy. Over half of students believed that an anonymously posted Facebook video, actually shot in Russia, provided *strong* evidence of U.S. voter fraud. Students displayed a troubling tendency to remain on a website and take it at face value. Over 96% never learned about ties between a climate change website and the fossil fuel industry. Instead of investigating the group behind the site, students were duped by weak signs of credibility: the website's "look", its top-level domain, or the content on its About page. Each of these features is ludicrously easy to game.

All students would benefit from carefully designed, thoughtfully delivered, and rigorously evaluated curriculum. At the same time, our findings suggest that the students most underserved by our nation's schools are similarly underserved in the realm of access to quality materials for evaluating digital sources. Socioeconomic status affects educational outcomes in the United States (cf. Mandara et al., 2009; Reardon et al., 2015) and indicators of socioeconomic status were predictors of scores here, too. After controlling for other variables, eligibility for the National School Lunch Program was negatively associated with student scores; higher maternal educational attainment was associated with higher scores. A pernicious racial opportunity gap persists in American schools (cf. Howard, 2019; Reardon et al., 2015), and our findings suggest it was at play as well.

It would be a mistake to see our results as an indictment of students. Presented with something to read, they mostly did what they've been taught. They read. Unless students possessed deep subject matter knowledge, a site like co2science.org—studded with citations to scientific articles, “major reports”, and lists of awards—looked, well, pretty good. The few students who learned that the website was backed by corporate interests did so not because they stayed on the site to engage in “close reading” (Common Core State Standards Initiative, n.d.). They succeeded because they didn't.

When the Internet is how we become informed citizens, traditional ways of reading are not merely ineffective (Gigerenzer & Gassmaier, 2011; Hertwig & Engel, 2016; Kozyreva et al., 2020). They're dangerous (Caulfield, 2018; Warzel, 2021). As far back as 1971, Herbert Simon presciently noted that information overload results in a scarcity of attention. Today, when advertisers, corporations, lobbyists, clickbait sites, conspiracy theorists, hate groups, and foreign governments work overtime to hijack attention, often the wisest thing to do is *not* to read, but to preserve attention by practicing *strategic ignoring* (Wineburg & McGrew, 2019). Under conditions of limited attention, the most crucial decision to make is where to allocate it. Above all else, lateral reading is an act of *attention conservation*. One uses the awesome powers of the Internet to quickly decide if an unfamiliar group, organization, or claim merits sustained attention. The few students who left a website to navigate the open Internet found answers waiting for them at the ready.

We have emphasized the need to leave a site to get an accurate read on it. Students would obviously benefit by mastering a host of other skills: choosing effective search terms, conducting reverse image searches, checking archived versions of webpages in the Internet Archive, and so on. Nor have we addressed the thorny issue of motivated reasoning, which renders the most

sophisticated searches moot if the searcher's mind is made up from the beginning. But we need to start somewhere. Unlike learning how to apply Boolean operators or decode a video's metadata, the basics of lateral reading are straightforward. Students open new tabs on their browsers and enter search terms, something they already know how to do. But they won't do so unless they understand, in deep and compelling ways, that the Web plays by a different set of rules from the vetted texts provided to them in school.

A much greater investment in research and development is needed to prepare students to navigate treacherous digital terrain. Small-scale but promising studies suggest that it is possible to improve students' digital savvy (Breakstone et al., 2021; Brodsky et al., 2019; Kohnen et al., 2020; McGrew, 2020; McGrew et al., 2019). Given the threat to civic and public health posed by digital illiteracy, these preliminary interventions need to be taken up at scale to identify approaches that can be widely adopted. Investigating how teachers across subject areas and grade levels integrate these skills into existing instruction will be critical. It's doubtful that students will become skilled consumers of digital content if instruction is restricted to a single course or one-off workshop.

Conclusion

Across regions and demographic groups, students in this study fell short. Their credulity makes them—along with the rest of us—easy prey for groups that seek to deceive, mislead, and manipulate. The toxic effects of disinformation have chipped away at the foundations of democracy the world over (Diamond, 2020; Levitsky & Ziblatt, 2018).

In October 1957, a whirling orbital ball known as Sputnik roused Americans from their

slumber. It led to the passage of the National Defense Education Act, a massive federal investment in American education. Today, the threat posed to national security by a digitally credulous citizenry warrants a similarly urgent response.

Anything less guarantees the continued erosion of democratic society.

Acknowledgements

This research was supported by the William and Flora Hewlett Foundation (Grant No. 2017-5939, Sam Wineburg, Principal Investigator). The content is solely the responsibility of the authors and does not necessarily represent the views of the William and Flora Hewlett Foundation.

Sarah McGrew and Teresa Ortega were integral to the research and development of Civic Online Reasoning assessments. This study would not have happened without Marc Chun's encouragement and support. Rex Long and Sarah Narvaiz provided valuable research assistance.

References

- Ashley, S., Maksl, A., & Craft, S. (2013). Developing a news media literacy scale. *Journalism & Mass Communication Educator*, 68(1), 7–21. <https://doi.org/10.1177/1077695812469802>
- Bartlett, J., & Miller, C. (2011). *Truth, lies and the Internet*. Demos.
https://www.demos.co.uk/files/Truth_-_web.pdf
- Barzilai, S., & Zohar, A. (2012). Epistemic thinking in action: Evaluating and integrating online sources. *Cognition and Instruction*, 30(1), 39–85.
<https://doi.org/10.1080/07370008.2011.636495>
- Breakstone, J., McGrew, S., Smith, M., Ortega, T., & Wineburg, S. (2018). Why we need a new approach to teaching digital literacy. *Phi Delta Kappan*, 99(6), 27-32.
<https://doi.org/10.1177/0031721718762419>
- Breakstone, J., Smith, M., Connors, P., Ortega, T., Kerr, D., & Wineburg, S. (2021). Lateral reading: College students learn to critically evaluate internet sources in an online course. *Harvard Kennedy School Misinformation Review*, 2(1), 1-17.
<https://doi.org/10.37016/mr-2020-56>
- Brodsky, J., Brooks, P. J., Scimeca, D., Todorova, R., Galati, P., Batson, M., Grosso, R., Matthews, M., Miller, V., Tachiera, T., & Caulfield, M. (2019, October 3-5). *Teaching college students the four moves of expert fact-checkers* [Paper presentation]. Technology, Mind, & Society, an American Psychological Association Conference, Washington, DC, United States.
- Caulfield, M. (2016, December 19). *Yes, digital literacy. But which one?* Hapgood.
<https://hapgood.us/2016/12/19/yes-digital-literacy-but-which-one/>

Caulfield, M. (2018, December 19). *Recalibrating our approach to misinformation*. EdSurge.

<https://www.edsurge.com/news/2018-12-19-recalibrating-our-approach-to-misinformation>

Coiro, J., Coscarelli, C., Maykel, C., & Forzani, E. (2015). Investigating criteria that seventh graders use to evaluate the quality of online information. *Journal of Adolescent and Adult Literacy*, 59(3), 287–297. <https://doi.org/10.1002/jaal.448>

Common Core State Standards Initiative. (n.d.). *English language arts standards*.

<http://www.corestandards.org/ELA-Literacy/CCRA/R/>

Dautrich, K. (2018). *The future of the First Amendment: 2018 survey of high school students and teachers*. John S. and James L. Knight Foundation.

<https://knightfoundation.org/reports/the-future-of-the-first-amendment-2018/>

Diamond, L. (2020). *Ill winds: Saving democracy from Russian rage, Chinese ambition, and American complacency*. Penguin.

Gasser, U., Cortesi, S., Malik, M., & Lee, A. (2012). *Youth and digital media: From credibility to information quality*. Berkman Klein Center for Internet & Society.

Gigerenzer, G., & Gaissmaier, W. (2011). Heuristic decision making. *Annual Review of Psychology*, 65, 451–482. <https://doi.org/10.1146/annurev-psych-120709-145346>

Hargittai, E., Fullerton, L., Menchen-Trevino, E., & Thomas, K. Y. (2010). Trust online: Young adults' evaluation of Web content. *International Journal of Communication*, 4, 468–494.

Hertwig, R., & Engel, C. (2016). Homo ignorans: Deliberately choosing not to know. *Perspectives on Psychological Science*, 11(3), 359–372. <https://doi.org/10.1177/1745691616635594>

Hobbs, R. (2010). *Digital and media literacy: A plan of action*. The Aspen Institute.

https://assets.aspeninstitute.org/content/uploads/2010/11/Digital_and_Media_Literacy.pdf

- Howard, T. C. (2019). *Why race and culture matter in schools: Closing the achievement gap in America's classrooms*. Teachers College Press.
- Kahne, J., Lee, N., & Feezell, J. T. (2012). Digital media literacy education and online civic and political participation. *International Journal of Communication*, 6, 1–24.
- Kohnen, A. M., Mertens, G. E., & Boehm, S. M. (2020). Can middle schoolers learn to read the web like experts? Possibilities and limits of a strategy-based intervention. *Journal of Media Literacy Education*, 12(2), 64-79. <https://doi.org/10.23860/JMLE-2020-12-2-6>
- Kozyreva, A., Lewandowsky, S., & Hertwig, R. (2020). Citizens versus the Internet: Confronting digital challenges with cognitive tools. *Psychological Science in the Public Interest*, 21(3), 103-156. <https://doi.org/10.1177/1529100620946707>
- Leu, D. J., Coiro, J., Kulikowich, J. M., & Cui, W. (2012, November 28-December 1). *Using the psychometric characteristics of multiple-choice, open Internet, and closed (simulated) Internet formats to refine the development of online research and comprehension assessments in science: Year three of the ORCA project* [Paper presentation]. Literacy Research Association 62nd Annual Conference, San Diego, CA, United States.
- Levitsky, S., & Ziblatt, D. (2018). *How democracies die*. Penguin.
- Lurie, E., & Mustafaraj, E. (2018). Investigating the effects of Google's search engine result page in evaluating the credibility of online news sources. *Proceedings of the 10th ACM Conference on Web Science*. ACM Digital Library. <https://doi.org/10.1145/3201064.3201095>
- Lynch, M. P. (2017). *An Internet of us: Knowing more and understanding less in an age of big data*. W.W. Norton.

- Mandara, J., Varner, F., Greene, N., & Richman, S. (2009). Intergenerational family predictors of the Black–White achievement gap. *Journal of Educational Psychology, 101*(4), 867–878. <https://doi.org/10.1037/a0016644>
- McGrew, S. (2020). Learning to evaluate: An intervention in civic online reasoning. *Computers & Education, 145*, 1-13. <https://doi.org/10.1016/j.compedu.2019.103711>
- McGrew, S., Breakstone, J., Ortega, T., Smith, M., & Wineburg, S. (2018). Can students evaluate online sources? Learning from assessments of civic online reasoning. *Theory and Research in Social Education, 46*(2), 165–193. <https://doi.org/10.1080/00933104.2017.1416320>
- McGrew, S., Smith, M., Breakstone, J., Ortega, T., & Wineburg, S. (2019). Improving students' web savvy: An intervention study. *British Journal of Educational Psychology, 89*(3), 485-500. <https://doi.org/10.1111/bjep.12279>
- Metzger, M. J., Flanagin, A. J., & Medders, R. B. (2010). Social and heuristic approaches to credibility evaluation online. *Journal of Communication, 60*(3), 413–439. <https://doi.org/10.1111/j.1460-2466.2010.01488.x>
- Mihailidis, P., & Thevenin, B. (2013). Media literacy as a core competency for engaged citizenship in participatory democracy. *American Behavioral Scientist, 57*(11), 1611–1622. <https://doi.org/10.1177/0002764213489015>
- National Center for Education Statistics. (n.d.-a). *Information on the Common Core of Data*. <https://nces.ed.gov/ccd/files.asp>
- National Center for Education Statistics. (n.d.-b). *Collecting race and ethnicity data from students and staff using the new categories*. <https://nces.ed.gov/ipeds/report-your-data/race-ethnicity-collecting-data-for-reporting-purposes>

National Center for Education Statistics. (n.d.-c). *NAEP is the leader in large-scale assessment*.

https://nces.ed.gov/nationsreportcard/about/leader_assessments.aspx

Pan, B., Hembrooke, H., Joachims, T., Lorigo, L., Gay, G., & Granka, L. (2007). In Google we trust: Users' decisions on rank, position, and relevance. *Journal of Computer-Mediated Communication*, 12(3), 801–823. <https://doi.org/10.1111/j.1083-6101.2007.00351.x>

Reardon, S. F., Robinson-Cimpian, J. P., & Weathers, E. S. (2015). Patterns and trends in racial/ethnic and socioeconomic academic achievement gaps. In H. F. Ladd & M. E. Goertz (Eds.), *Handbook of research in education finance and policy* (2nd ed., pp. 491-509). Routledge.

Reardon, S. F., & Townsend, J. B. (2018). SEG: Stata module to compute multiple-group diversity and segregation indices [Statistical software module]. EconPapers. <https://econpapers.repec.org/software/bocbocode/s375001.htm>

Rideout, V., & Robb, M. B. (2019). *The Common Sense census: Media use by tweens and teens, 2019*. Common Sense Media. <https://www.commonsensemedia.org/research/the-common-sense-census-media-use-by-tweens-and-teens-2019>

Simon, H. A. (1971). Designing organizations for an information-rich world. In M. Greenberger (Ed.), *Computers, communications, and the public interest* (pp. 38-72). The Johns Hopkins Press. <https://digitalcollections.library.cmu.edu/awweb/awarchive?type=file&item=33748>

Sullivan, M. C. (2019). Why librarians can't fight fake news. *Journal of Librarianship and Information Science*, 51(4), 1146-1156. <https://doi.org/10.1177/0961000618764258>

- Walraven, A., Brand-Gruwel, S., & Boshuizen, H. P. A. (2009). How students evaluate information and sources when searching the World Wide Web for information. *Computers & Education*, 52(1), 234–246. <https://doi.org/10.1016/j.compedu.2008.08.003>
- Warzel, C. (2021, February 18). Don't go down the rabbit hole. *The New York Times*. <https://www.nytimes.com/2021/02/18/opinion/fake-news-media-attention.html>
- Wineburg, S., Breakstone, J., Ziv, N., & Smith, M. (2020). *Educating for misunderstanding: How approaches to teaching digital literacy make students susceptible to scammers, rogues, bad actors, and hate mongers* (Working Paper A-21322), Stanford History Education Group. <https://purl.stanford.edu/mf412bt5333>
- Wineburg, S., & McGrew, S. (2017). *Lateral reading: Reading less and learning more when evaluating digital information* (Working Paper No. 2017-A1), Stanford History Education Group. <http://dx.doi.org/10.2139/ssrn.3048994>
- Wineburg, S., & McGrew, S. (2019). Lateral reading and the nature of expertise: Reading less and learning more when evaluating digital information. *Teachers College Record*, 121(11), 1-40.
- Wineburg, S., McGrew, S., Breakstone, J., & Ortega, T. (2016). *Evaluating information: The cornerstone of Civic Online Reasoning* (Working Paper No. 2016-A1), Stanford History Education Group. <http://purl.stanford.edu/fv751yt5934>
- Wineburg, S., & Ziv, N. (2019, December 5). The meaninglessness of the .org domain. *The New York Times*. <https://www.nytimes.com/2019/12/05/opinion/dot-org-domain.html>
- Wineburg, S., & Ziv, N. (2020, November 6). Why can't a generation that grew up online spot the misinformation in front of them? *Los Angeles Times*.

<https://www.latimes.com/opinion/story/2020-11-06/colleges-students-recognize-misinformation>

Appendix A

Description of Tasks on COR Assessment

Item	COR Question(s) Addressed	Description
1. Evaluating Video Evidence	What's the evidence? Who's behind the information?	Evaluate whether a video posted on Facebook is good evidence of voter fraud.
2. Webpage Comparison	Who's behind the information?	Explain which of two webpages is a better source of information on gun control.
3. Article Evaluation	What do other sources say? Who's behind the information?	Using any online sources, explain whether a website is a reliable source of information about global warming.
4. Claims on Social Media 1	What's the evidence? Who's behind the information?	Explain why a social media post is a useful source of information about background checks for firearms.
5. Claims on Social Media 2	What's the evidence? Who's behind the information?	Explain how a social media post about background checks might not be a useful source of information.
6. Home Page Analysis	Who's behind the information?	Explain whether tiles on the homepage of a website are advertisements or news stories.

Appendix B

Students' Online Reasoning Study

You are invited to participate in a research project that looks at how people evaluate online sources. Findings from this project will help us create resources to teach students how to navigate and evaluate information on the Internet.

Completing the tasks will take about 30-40 minutes. Your answers are completely anonymous. Your answers will not affect your grade in this course in any way.

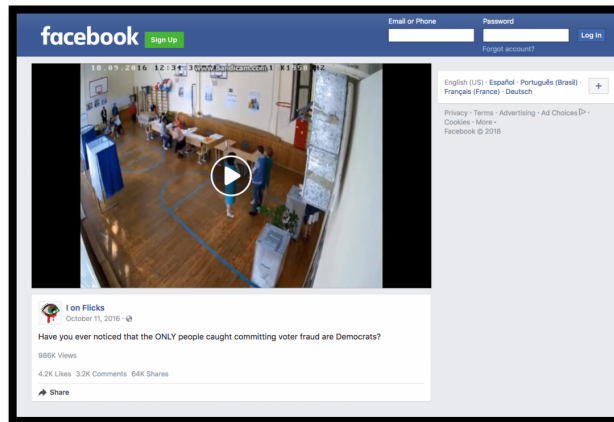
On the following pages you will be presented with five tasks. Please keep the following in mind as you complete each one:

- You may open a new tab and complete an Internet search to help you answer any of the questions.
- Once you advance, you will not be able to return to previous pages. Your browser's back button won't work either.

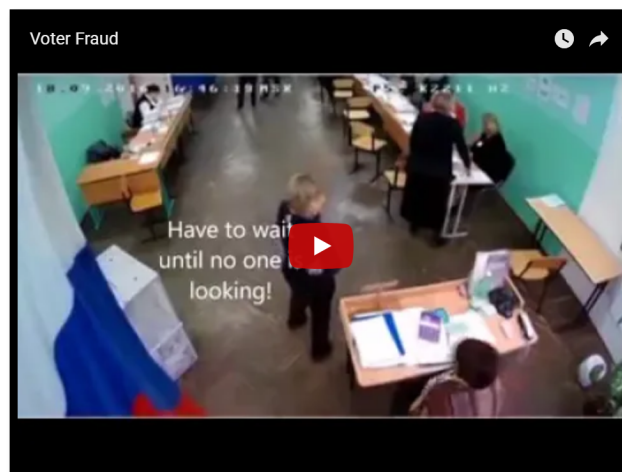
Thank you for participating!

Task 1 of 5

Below is a screenshot of a Facebook post from October 2016. Please watch the video posted below the screenshot.



You can watch the video from the post here: (Note, there is no audio.)



Question: Does the Facebook post provide strong evidence of voter fraud during the 2016 Democratic primary election? (The primary was the election to decide who would be the Democratic candidate for president.)

- Yes
- No

Explain your reasoning.

Were you able to view the video above? (Note: The video does not have sound.)

- Yes
- No

Task 2 of 5

Imagine you are doing research on gun control and you find the following webpages. Because webpages can change frequently, these links will take you to archived versions of the articles. Please do not consider that in your evaluations.

- <http://web.archive.org/web/20161205141632/http://people.duke.edu/~gnsmith/articles/myths.htm>
- http://web.archive.org/web/20170222133426/https://en.wikipedia.org/wiki/Gun_politics_in_the_United_States

Question: Which of these two webpages do you think is a better place to start your research?

- <http://people.duke.edu/~gnsmith/articles/myths.htm>
- https://en.wikipedia.org/wiki/Gun_politics_in_the_United_States

Explain your choice. Refer to both webpages in your answer.

Were you able to access both links about gun control?

- Yes
- No

Task 3 of 5

Visit the following webpage: <http://www.co2science.org/>

Question: Is this a reliable source of information about global warming? You may use any information on this website, or you can open a new tab and do an Internet search if you want.

- Yes
- No

Explain your answer, citing evidence from the webpages you used. Be sure to provide the URLs to the webpages you cite.

Were you able to access the above link about climate science?

- Yes
- No

Task 4 of 5

The following tweet appears in your Twitter feed (please click on the link to view):

<https://twitter.com/MoveOn/status/666772893846675456?lang=en>

Question: Why might this tweet be a useful source about NRA members' opinions on background checks? List any sources you used to make your decision.

Question: Why might this tweet not be a useful source about NRA members' opinions on background checks? List any sources you used to make your decision.

Were you able to view the tweet above?

- Yes
- No

Task 5 of 5

Here is the home page of Slate.com. Some of the things that appear on Slate.com are news stories, and others are advertisements.

The screenshot shows the Slate.com homepage. At the top, there's a banner for 'We know you've got a story.' with a 'GOTHAM WRITERS' logo and a 'SAVE \$20' offer. Below this, there are three main sections:

- Section 1 (Red circle #1):** An advertisement for 'GOTHAM WRITERS' with a 'SAVE \$20' offer and a 'LIMITED TIME ONLY' tag.
- Section 2 (Red circle #2):** An article titled 'Should California Stop Growing Almonds?' by Eric Holthaus. The article features a photo of almonds and a quote: 'The nut has been vilified for drinking up the state's water supply. It doesn't deserve such a bad rap.'
- Section 3 (Red circle #3):** A sponsored content article titled 'The Real Reasons Women Don't Go Into Tech' by Laura Bradley and Marie Lindemann. It features a small chart showing percentages: 37% and 12%.

On the right side of the page, there's a 'Slate' logo and a 'MOST RECENT' section with a 'SEE ALL' link. Below this, there are three recent articles:

- 24M AGO - JORDAN WEISSMAN - 1M TO READ: 'Forget Steak and Seafood: Here's How Welfare Recipients Actually Spend Their Money'
- 30M AGO - BEN MATHIS-LILLEY: 'Buckingham Palace Guard Falls Over (Video)'
- 45M AGO - L. BRADLEY & A.M. LINDEMANN - 2M TO READ: 'When is Cheryl's Birthday? Solving a Logic Problem That Quickly Spread Around the World.'

At the bottom right, there's an advertisement for 'slow watches' featuring a photo of a watch and the text 'right here' and 'slow watches.com'.

Question: Is #1 an advertisement?

- Yes
- No

Explain your reasoning.

Question: Is #2 an advertisement?

- Yes
- No

Explain your reasoning.

Question: Is #3 an advertisement?

- Yes
- No

Explain your reasoning.

Additional Questions

On a scale of 0-10, how seriously did you take these tasks? "0" indicates that you did not take the tasks seriously at all. "5" indicates that you took the tasks fairly seriously. "10" indicates that you took the tasks very seriously.

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

Thank you for completing this exercise! Please take a few minutes to answer the questions below. Your answers to these questions help us learn about the students who completed the exercises. All of your responses are still anonymous.

Please select your school district.

Please select your school.

In which class you are completing this survey?	Is this class an Honors or IB class?	What grade are you in?
- US/American History I - US/American History II - World History - AP US History - AP US Government & Politics - Other (please enter class name)	- Yes, Honors - Yes, IB - No	8th grade or below 9th 10th 11th 12th

How old are you?	What is your gender?	Are you Hispanic or Latino?
12 or younger 13 14 15 16 17 18 19 20 or older	- Female - Male - Non-binary - Prefer to self-describe - Prefer not to say	- Yes - No - Prefer not to say

What is your race? Mark one of more boxes
- American Indian or Alaska Native - Asian - Black or African American - Native Hawaiian or Other Pacific Islander - White

What is the highest degree or level of school that your mother completed?
- Less than high school - High school - Some college - Bachelor's degree - Advanced degree (e.g., MA, JD, MD, PhD).

- Other
- Prefer not to say

- I don't know
- Prefer not to say

What is the language(s) most used in your home? Drag and drop up to three languages from the left column to the right box. Then rank them in order of how much they're used in your home.

- English
- Arabic
- Burmese
- Cambodian
- Cantonese
- Cape Verdean Creole
- French
- Greek
- Hmong
- Haitian-Creole
- Italian
- Korean
- Mandarin
- Portuguese
- Russian
- Somali
- Spanish
- Toishanese
- Vietnamese
- Other (Please specify):
- Prefer not to say

Most used language(s) in your home (select and rank up to 3)

What is the zip code where you live? (If you live in multiple zip codes, choose the one in which you spend more time.)

Are you eligible for free or reduced-price lunch at school?

- Yes
- No
- I don't know.
- Prefer not to say

Which best describes the grades you usually earn on your report card?

- Mostly A's

- Mostly A's and B's
- Mostly B's
- Mostly B's and C's
- Mostly C's
- Mostly C's and D's
- Mostly D's
- Prefer not to say

Appendix C

Task 1: Evaluating Evidence Rubric

This assessment asks students to determine whether a video posted on Facebook provides strong evidence of voter fraud in the 2016 election. Successful students will conclude that the video is not strong evidence and provide a clear explanation why it isn't. Successful explanations can focus on a variety of factors, including: (1) there is no evidence that the video was actually shot where the poster claims it was; (2) nothing is known about the identity of the person who posted the video (I on Flicks) and they are likely not a reliable source of information; (3) after doing research on the video, it is clear that the video is not what the poster claims it is; or (4) after researching the identity of the person who posted the video, it is clear that "I on Flicks" may not be a reliable source of information.

(2) MASTERY	Student rejects the video as strong evidence and does at least one of the following: • Questions who posted the video • Questions the source/location of the video • Investigates who posted the video and concludes they are unreliable • Investigates the video and determines the recordings are not from the 2016 Democratic primary
(1) EMERGING	Student questions the content of the video rather than the source, implying or stating that an unedited video would be automatically reliable OR Student questions the source of the post/video, but also does one of the following: • Waffles/straddles the fence, both accepting the video as good evidence but also raising relevant concerns • States incorrect reasoning for why the post/video is unreliable
(0) BEGINNING	Student rejects the video but uses completely incorrect or incoherent reasoning. OR Student concludes the post <u>does</u> provide strong evidence.

Appendix D

Regression Model Demographic Variables

Race/ethnicity: We included the composite race/ethnicity variable in the multivariate models. Students who did not answer the question or who answered “prefer not to say” were not included.

Gender: We included a categorical measure of gender with three categories: female, male, and a third category that combined nonbinary and “prefer to self-describe” response options. Individuals who did not answer the question or answered “prefer not to say” were not included in the multivariate models.

Free/reduced lunch status: FRL status was a binary indicator (eligible/participating versus not eligible/not-participating). Students who did not answer the question or who responded “I don’t know” or “prefer not to say” were not included in the multivariate models.

Mother’s education: Student reports on their mother’s highest level of education were coded 1 (less than a high school education) to 5 (advanced degree). Students who did not answer the question or who responded “I don’t know” or “prefer not to say” were not included in the multivariate models.

Home language: The assessments were in English, so students for whom English was not a first language may have had more difficulty interpreting the sources and conveying their thinking in English. As a result, we included a categorical measure of home language (English versus not English) based on the students’ response about the language(s) spoken at home. Students who did not answer the question were not included in the multivariate models.

Appendix E

Descriptive Results Tables

Table E1

Percent Scoring Emerging or Mastery by Locale for Tasks 1 to 5

Locale	N	Evaluating Evidence %	Webpage Comparison %	Website Evaluation %	Claims on Social Media 1 %	Claims on Social Media 2 %	Home Page Analysis %
Rural	820	20.9%	4.2%	2.6%	5.0%	2.3%	23.1%
Suburban	1,659	20.9%	7.9%	2.7%	6.8%	3.7%	31.0%
Urban	967	27.3%	11.2%	4.6%	10.6%	9.3%	39.4%
Total	3,446	3,119	2,678	2,994	2,431	2,431	2,969

Table E2

Percent Scoring Emerging or Mastery by Region for Tasks 1 to 5

Region	N	Evaluating Evidence %	Webpage Comparison %	Website Evaluation %	Claims on Social Media 1 %	Claims on Social Media 2 %	Home Page Analysis %
Midwest	852	16.4%	4.5%	1.7%	4.0%	2.5%	26.9%
Northeast	732	15.3%	5.2%	2.3%	4.2%	1.1%	22.4%
South	1,084	27.6%	10.4%	4.2%	10.0%	7.2%	37.1%
West	778	29.2%	10.1%	3.9%	5.4%	3.7%	36.4%
Total	3,446	3,119	2,678	2,994	2,431	2,431	2,969

Table E3

Percent Scoring Emerging or Mastery by Grade for Tasks 1 to 5

Grade	N	Evaluating Evidence %	Webpage Comparison %	Website Evaluation %	Claims on Social Media 1 %	Claims on Social Media 2 %	Home Page Analysis %
8th Grade	17	14.3%	0.0%	0.0%	0.0%	0.0%	13.3%
9th Grade	820	15.1%	5.2%	1.3%	4.2%	1.5%	24.4%
10th Grade	491	19.8%	5.8%	1.5%	6.0%	4.3%	25.7%
11th Grade	867	26.7%	9.6%	3.6%	7.2%	5.5%	36.1%
12th Grade	761	29.2%	11.8%	5.6%	12.1%	8.0%	38.8%
Missing	490	20.0%	4.9%	3.8%	5.5%	1.1%	10.9%
Total (N)	3,446	3,119	2,678	2,994	2,431	2,431	2,969

Table E4

Percent Scoring Emerging or Mastery by Race/Ethnicity and Task

Race/Ethnicity	N	Eval Evidence %	Webpage Comparison %	Website Evaluation %	Claims on Social Media 1 %	Claims on Social Media 2 %	Home Page Analysis %
Asian/Pacific Islander	112	33.6%	22.8%	5.4%	16.7%	7.2%	43.8%
American Indian/Alaska Native	29	17.9%	3.8%	0.0%	11.5%	3.8%	31.0%
Black or African American	306	11.0%	2.8%	0.0%	2.2%	1.3%	17.6%
Hispanic	698	17.6%	3.9%	0.9%	3.9%	3.2%	24.7%
Other	42	13.5%	6.7%	2.7%	6.5%	9.4%	31.0%
Two or more races	224	29.5%	9.0%	3.2%	6.5%	6.0%	32.6%
White	1,373	27.8%	11.5%	5.0%	9.7%	6.1%	38.3%
Prefer not to say	28	19.2%	10.0%	7.4%	8.7%	4.3%	37.0%
Missing	634	18.0%	3.6%	2.2%	3.6%	0.5%	17.0%
Total	3,446	3,119	2,678	2,994	2,431	2,431	2,969

Table E5

Percent Scoring Emerging or Mastery by Average Grades for Tasks 1 to 5

Grades	N	Evaluating Evidence %	Webpage Comparison %	Website Evaluation %	Claims on Social Media 1 %	Claims on Social Media 2 %	Home Page Analysis %
Mostly As	794	38.9%	14.7%	6.5%	13.8%	10.0%	43.1%
Mostly As and Bs	1,116	21.4%	6.6%	2.4%	6.4%	3.5%	32.1%
Mostly Bs	268	16.4%	6.9%	2.7%	4.4%	3.8%	24.8%
Mostly Bs and Cs	445	11.1%	3.7%	1.0%	2.8%	1.1%	21.1%
Mostly Cs	108	12.9%	3.5%	0.0%	2.8%	1.4%	24.5%
Mostly Cs and Ds	77	10.8%	10.2%	0.0%	1.8%	1.8%	23.4%
Mostly Ds	33	7.7%	0.0%	0.0%	0.0%	0.0%	26.7%
Prefer not to say	101	7.8%	3.6%	1.1%	1.4%	0.0%	20.0%
Missing	504	19.5%	5.0%	3.5%	4.9%	1.0%	10.7%
Total	3,446	3,119	2,678	2,994	2,431	2,431	2,969

Table E6

Percent Scoring Emerging or Mastery by Gender for Tasks 1 to 5

Gender	N	Evaluating Evidence %	Webpage Comparison %	Website Evaluation %	Claims on Social Media 1 %	Claims on Social Media 2 %	Home Page Analysis %
Female	1,443	24.5%	7.1%	2.7%	7.1%	4.3%	31.8%
Male	1,283	22.2%	10.2%	4.0%	7.8%	5.7%	32.0%
Non-binary	28	20.0%	5.0%	9.1%	0.0%	5.3%	46.2%
Prefer to self-describe	144	15.8%	1.8%	0.0%	5.3%	2.2%	25.4%
Prefer not to say	58	15.8%	1.8%	0.0%	5.3%	2.2%	25.4%
Missing	490	20.0%	4.9%	3.8%	5.5%	1.1%	10.9%
Total	3,446	3,119	2,678	2,994	2,431	2,431	2,969

Table E7

Percent Scoring Emerging or Mastery by National School Lunch Program Eligibility for Tasks 1 to 5

Eligible for School Lunch Program	N	Evaluating Evidence %	Webpage Comparison %	Website Evaluation %	Claims on Social Media 1 %	Claims on Social Media 2 %	Home Page Analysis %
Yes	850	16.0%	4.4%	0.5%	3.7%	0.9%	21.1%
No	1,475	29.7%	11.5%	5.3%	10.8%	8.6%	39.9%
I don't know	533	16.8%	6.1%	1.6%	4.2%	1.2%	28.9%
Prefer not to say	84	13.9%	2.7%	0.0%	3.3%	0.0%	16.0%
Missing	504	19.5%	5.0%	3.5%	4.9%	1.0%	10.7%
Total	3,446	3,119	2,678	2,994	2,431	2,431	2,969

Table E8

Percent Scoring Emerging or Mastery by Mother's Education for Tasks 1 to 5

Highest degree/level of education mother completed?	N	Evaluating Evidence %	Webpage Comparison %	Website Evaluation %	Claims on Social Media 1 %	Claims on Social Media 2 %	Home Page Analysis %
Less than high school	222	20.3%	5.4%	1.4%	5.9%	3.5%	25.6%
High school	506	14.5%	3.7%	0.4%	3.6%	1.3%	21.9%
Some college	460	22.8%	9.2%	2.3%	4.9%	3.3%	29.5%
Bachelors degree	736	28.1%	11.4%	4.5%	9.5%	7.2%	39.5%
Advanced degree	473	34.6%	13.4%	7.7%	15.0%	11.7%	43.6%
I don't know	411	17.4%	5.1%	1.3%	3.6%	0.3%	25.7%
Prefer not to say	148	8.1%	3.2%	0.7%	1.9%	0.0%	21.5%
Missing	490	20.0%	4.9%	3.8%	5.5%	1.1%	10.9%
Total	3,446	3,119	2,678	2,994	2,431	2,431	2,969

Table E9

Percent Scoring Emerging or Mastery by Home Language for Tasks 1 to 5

Home Language	<i>N</i>	Evaluating Evidence %	Webpage Comparison %	Website Evaluation %	Claims on Social Media 1 %	Claims on Social Media 2 %	Home Page Analysis %
English	2,425	24.8%	8.9%	3.6%	8.0%	5.4%	33.7%
Spanish	302	13.1%	2.6%	0.3%	2.1%	0.4%	21.4%
Other	219	17.8%	9.1%	2.0%	6.3%	3.2%	24.0%
Missing	500	19.5%	4.7%	3.6%	5.2%	1.1%	11.1%
Total	3,446	3,119	2,678	2,994	2,431	2,431	2,969

Appendix F

Distribution of the Composite Scores

Composite Score	<i>N</i>	%
0.0	1,974	58.0%
0.17 to 0.49	966	28.4%
0.5 to 0.99	351	10.3%
1.0 to 1.49	88	2.6%
1.5 to 1.99	10	0.29%
2.0	13	0.38%
Total	3,402 ^a	100%

^a Forty-four students in the analytic sample dropped out of the composite score analysis because the students indicated technical problems or provided nonsensical answers.